EXPERIMENT-3

1.COMMAND LINE INTERFACE

(i)CLI Python Code:

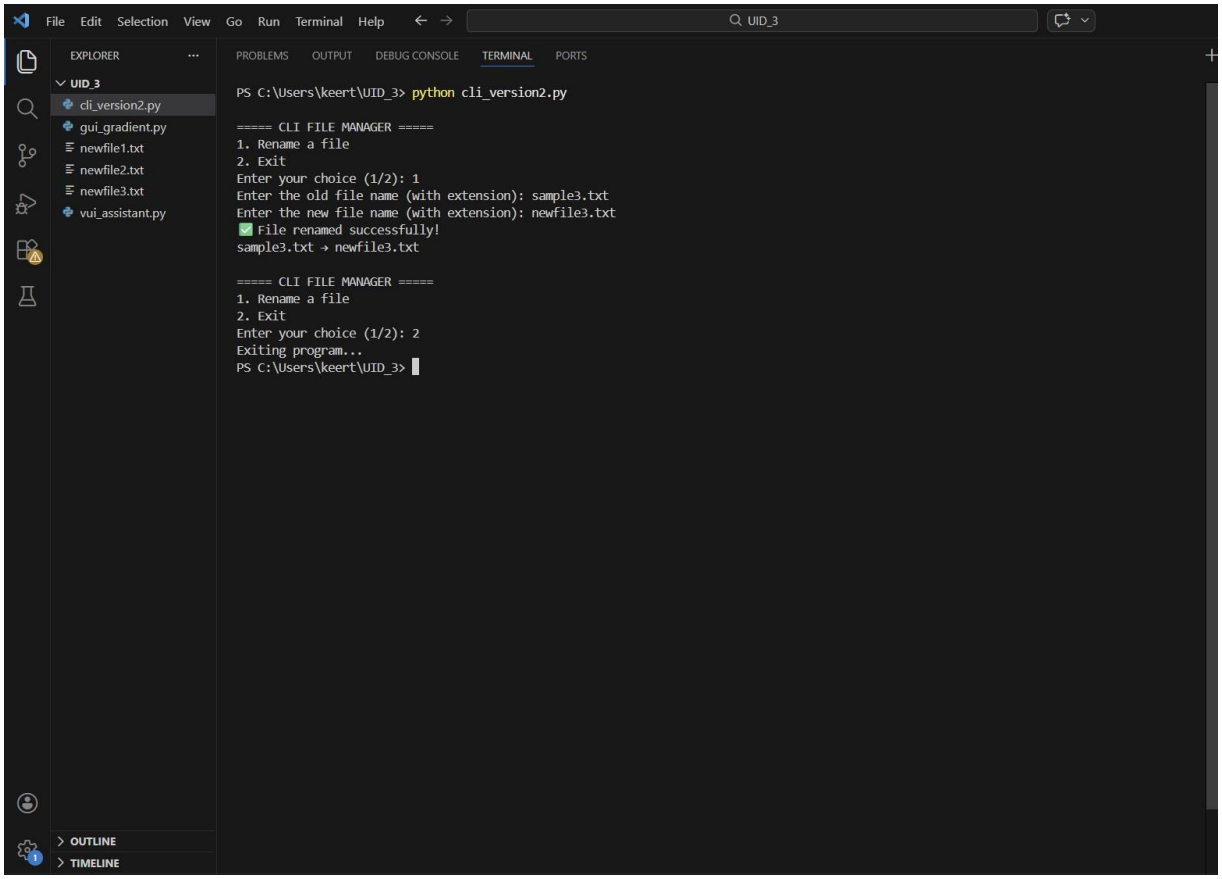
cli_app.py:

```
import os

def rename_file():

  old_name = input("Enter the old file name (with extension): ")

  new_name = input("Enter the new file name (with extension): ")

  try:

    if not os.path.isfile(old_name):

      print("File does not exist.")

      return

    if os.path.isfile(new_name):
              print("A file with new name already exists.")
              return
```

```
      os.rename(old_name, new_name)
      print(“File renamed successfully!”)
     print(f"{old_name} → {new_name}")

  except Exception as e:
   print("Error occurred: ", e)

def main():
while True:
    print("\n===== CLI FILE MANAGER =====")
    print("1. Rename a file")
    print("2. Exit")
    choice = input("Enter your choice (1/2): ")
    if choice=="1":
      rename_file
      ()
    elif choice=="2":
    print("Exiting program...")
    break
    else:
      print("Invalid choice. Try again.")

if __name__== "__main__":
```

```
main()
```

OUTPUT :-


File Edit Selection View Go Run Terminal Help
UID_3
EXPLORER
UID_3
cli_version2.py
gui_gradient.py
newfile1.txt
newfile2.txt
newfile3.txt
vui_assistant.py
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\keert\UID_3> python cli_version2.py
===== CLI FILE MANAGER =====
1. Rename a file
2. Exit
Enter your choice (1/2): 1
Enter the old file name (with extension): sample3.txt
Enter the new file name (with extension): newfile3.txt
✅ File renamed successfully!
sample3.txt → newfile3.txt
===== CLI FILE MANAGER =====
1. Rename a file
2. Exit
Enter your choice (1/2): 2
Exiting program...
PS C:\Users\keert\UID_3>

(ii)GUI (PYTHON CODE) :-

```
import tkinter as tk

from tkinter import
messagebox import os


def rename_file():
     old_name = old_entry.get()

      new_name = new_entry.get()
```

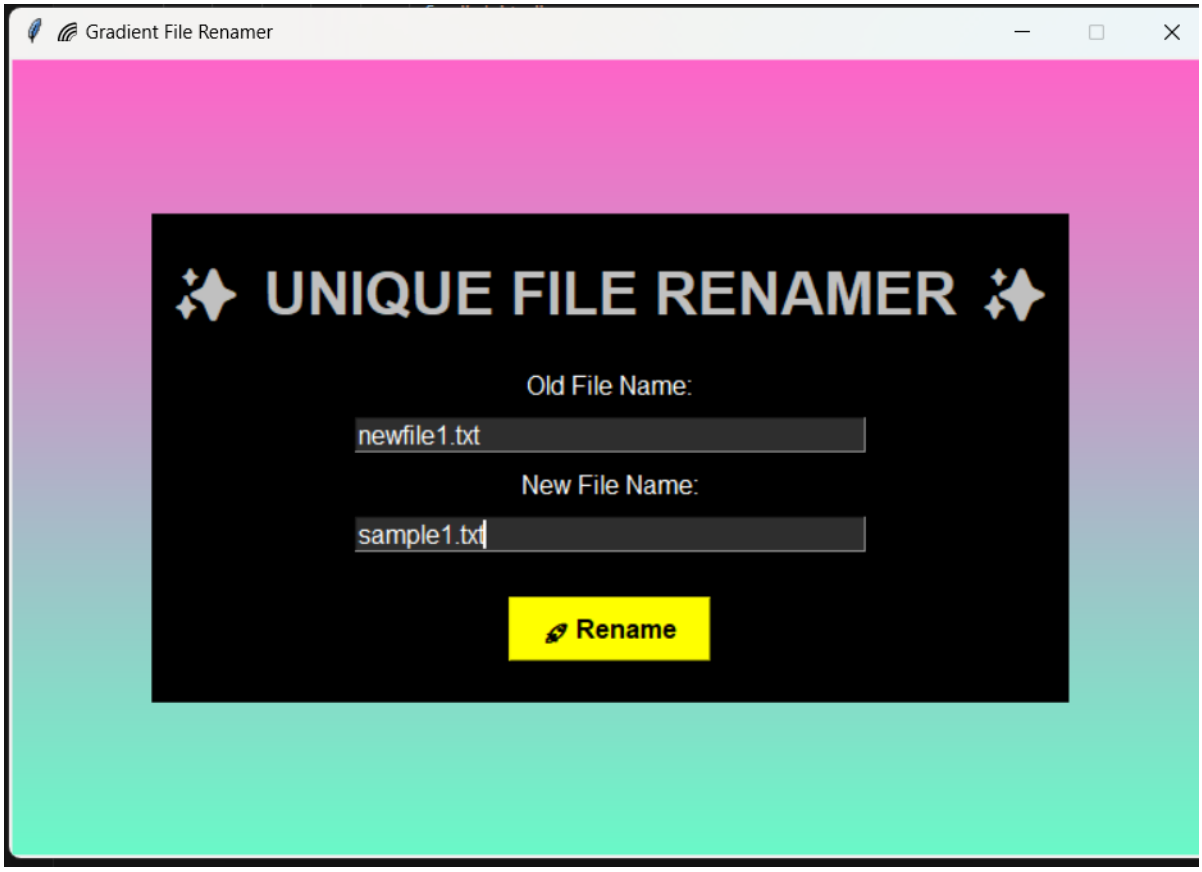
```
if old_name == "" or new_name == "":
    messagebox.showwarning("Input Error", "Please enter both
file names!")
    return

  if not os.path.isfile(old_name):
    messagebox.showerror("Error", "File not
    found!") return

  if os.path.isfile(new_name):
    messagebox.showwarning("Warning", "New file name already
    exists!") return

  os.rename(old_name, new_name)
  messagebox.showinfo("Success ✅ f"{old_name} ➜
  {new_name}") old_entry.delete(0, tk.END)
  new_entry.delete(0, tk.END)

root = tk.Tk()
root.title("Gradient file
renamer")
root.geometry("750x500")
root.resizable(False, False)
```

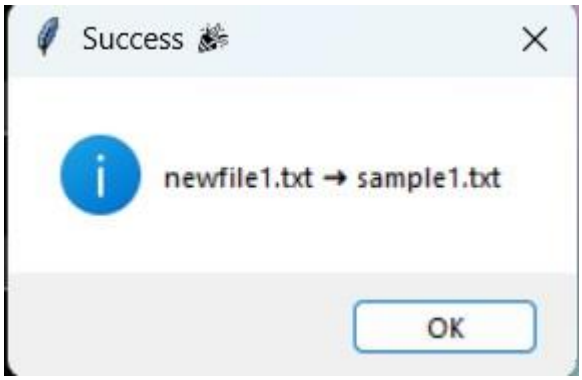
```
canvas = tk.Canvas(root, width=750, height=500)
canvas.pack(fill="both", expand=True)
for i in range(500):
  r = max(0, min(255, int(255 - i *
  0.3))) g = max(0, min(255, int(100
  + i * 0.3))) b = 200
  color = f'#{r:02x}{g:02x}{b:02x}'
  canvas.create_line(0, i, 750, i, fill=color)

frame = tk.Frame(root, bg="black", bd=5)
frame.place(relx=0.5, rely=0.5, anchor="center")

tk.Label(frame,
    text="UNIQUE FILE RENAMER",
    font=("Helvetica", 28,
    "bold"), bg="black",
    fg="#c0c0c0").pack(pady=20)

tk.Label(frame,
    text="Old File
    Name:",
    font=("Arial",
    12), bg="black",
    fg="white").pack()
```

```
old_entry = tk.Entry(frame,
         width=35,
         font=("Arial", 12),
         bg="#2e2e2e",
         fg="white",
         insertbackground="white")
old_entry.pack(pady=8)

tk.Label(frame,
    text="New File Name:",
    font=("Arial", 12),
    bg="black",
    fg="white").pack()

new_entry = tk.Entry(frame,
         width=35,
         font=("Arial", 12),
         bg="#2e2e2e",
         fg="white",
         insertbackground="white")
new_entry.pack(pady=8)

tk.Button(frame,
     text="Rename"
```

```
    font=("Arial", 12,
    "bold"), bg="yellow",
    fg="black",
    activebackground="orange"
    , padx=15,
    pady=5,
    command=rename_file).pack(pady=20)

root.mainloop()
```

OUTPUT :-





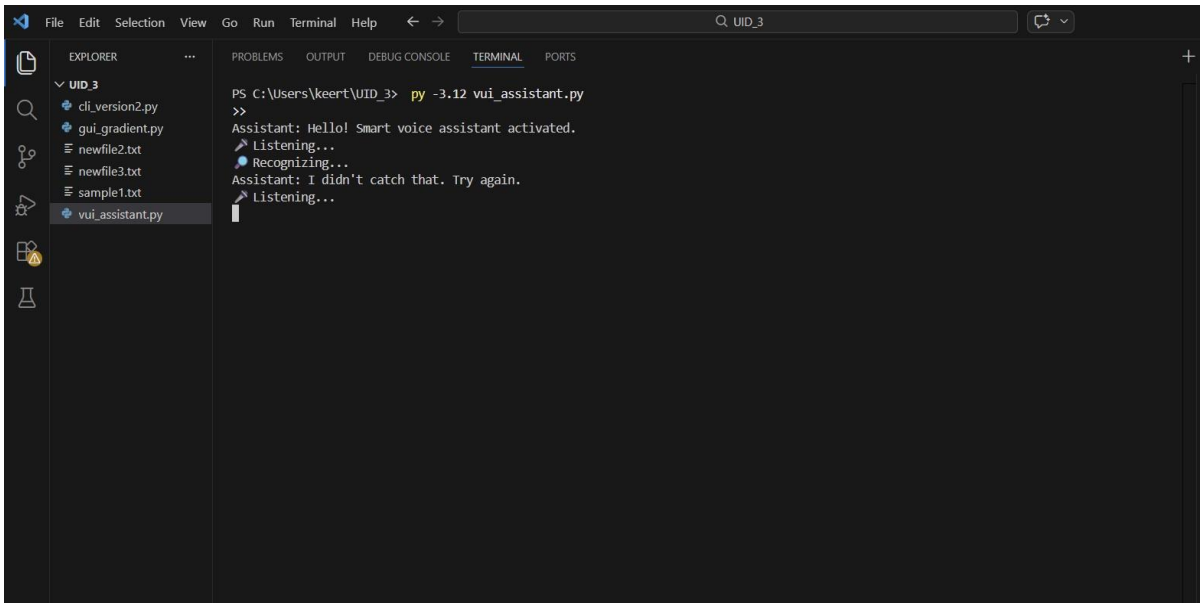
2. VUI (PYTHON CODE):-

```
import speech_recognition as sr
import pyttsx3
import datetime
import webbrowser

# Initialize recognizer C
speaker recognizer =
sr.Recognizer() engine =
pyttsx3.init()

def speak(text):
  print("Assistant:", text)
  engine.say(text)
  engine.runAndWait()

def listen():
```

```
  with sr.Microphone() as
    source: print("🎤
    Listening...")
    recognizer.adjust_for_ambient_noise(source,
    duration=1)
    audio = recognizer.listen(source)

  try:
    print("🔍 Recognizing...")
    command = recognizer.recognize_google(audio)
    print("You said:", command)
    return
  command.lower()
  except:
    speak("Sorry, I didn't
    understand.") return None

def main():
  speak("Hello! I am your voice assistant.")

  while True:
    command = listen()

    if command is None:
      continue
```

```
    if "hello" in command:
      speak("Hello boss!")

    elif "time" in command:
      current_time  =
      datetime.datetime.now().strftime("%I:%M
%p")
      speak(f"The time is {current_time}")

    elif "open youtube" in command:
      speak("Opening YouTube")
      webbrowser.open("https://youtube.com")

    elif "exit" in command or "stop" in command:
      speak("Goodbye!")
      break

if __name__ == "__main
  __": main()
```

OUTPUT :-


File Edit Selection View Go Run Terminal Help
UID_3
EXPLORER
UID_3
cli_version2.py
gui_gradient.py
newfile2.txt
newfile3.txt
sample1.txt
vui_assistant.py
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\keert\UID_3> py -3.12 vui_assistant.py
>>
Assistant: Hello! Smart voice assistant activated.
🎤 Listening...
🔎 Recognizing...
Assistant: I didn't catch that. Try again.
🎤 Listening...